AI Voice Emotion Analyzer

Introduction

The AI Voice Emotion Analyzer is a system designed to analyze audio input files and extract meaningful voice characteristics such as pitch, frequency, and energy, along with emotion prediction. The system now includes **Text-to-Speech (TTS) output generation**, allowing users to download a synthesized voice version of the analysis results.

The system accepts audio files in formats like .wav and .mp3, processes them using signal processing techniques, outputs both numerical values and interpreted results (e.g., High Pitch, Low Frequency, Calm Emotion), and **generates a new audio file with spoken analysis results**.

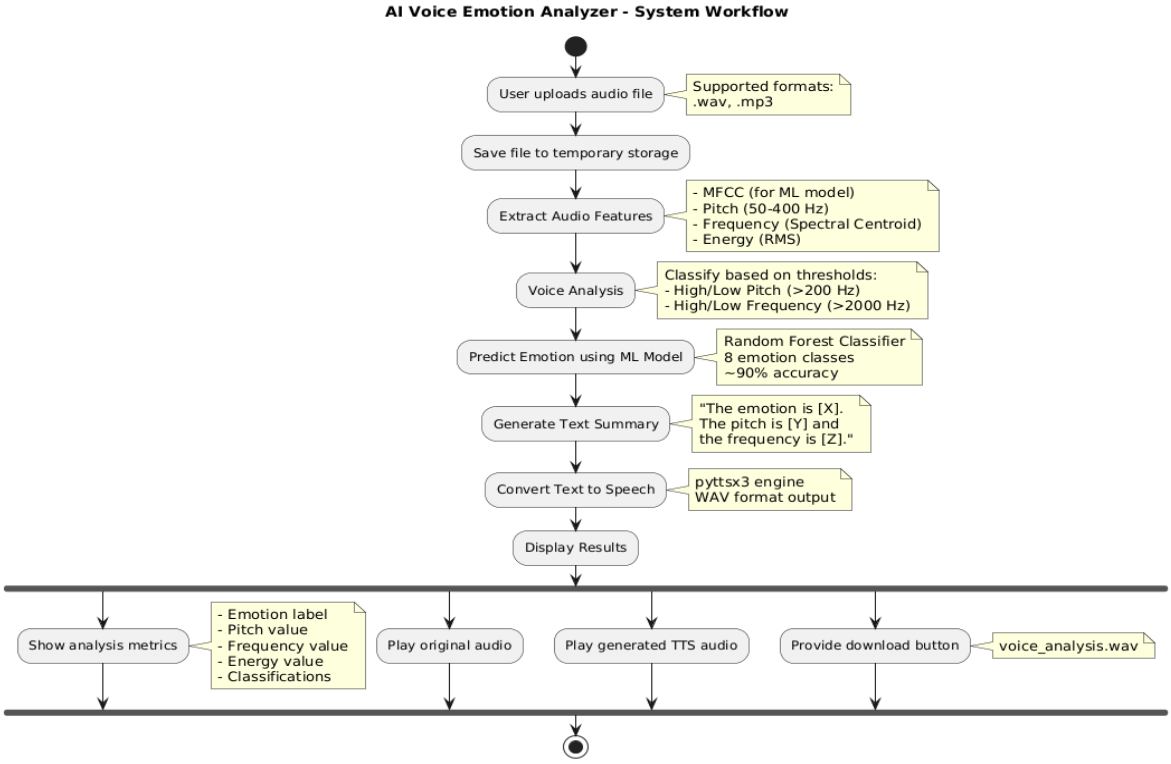
This project combines audio signal processing, machine learning, and speech synthesis to provide a complete voice analysis solution.

Objective

The main objectives of this project are:

- To analyze voice signals from audio files
- To extract key audio features (Pitch, Frequency, Energy)
- To classify voice characteristics into human-understandable categories
- To predict emotion using a trained machine learning model
- **To generate synthesized speech output based on the analysis**
- **To provide downloadable audio results**
- To provide a simple and interactive user interface

System Workflow

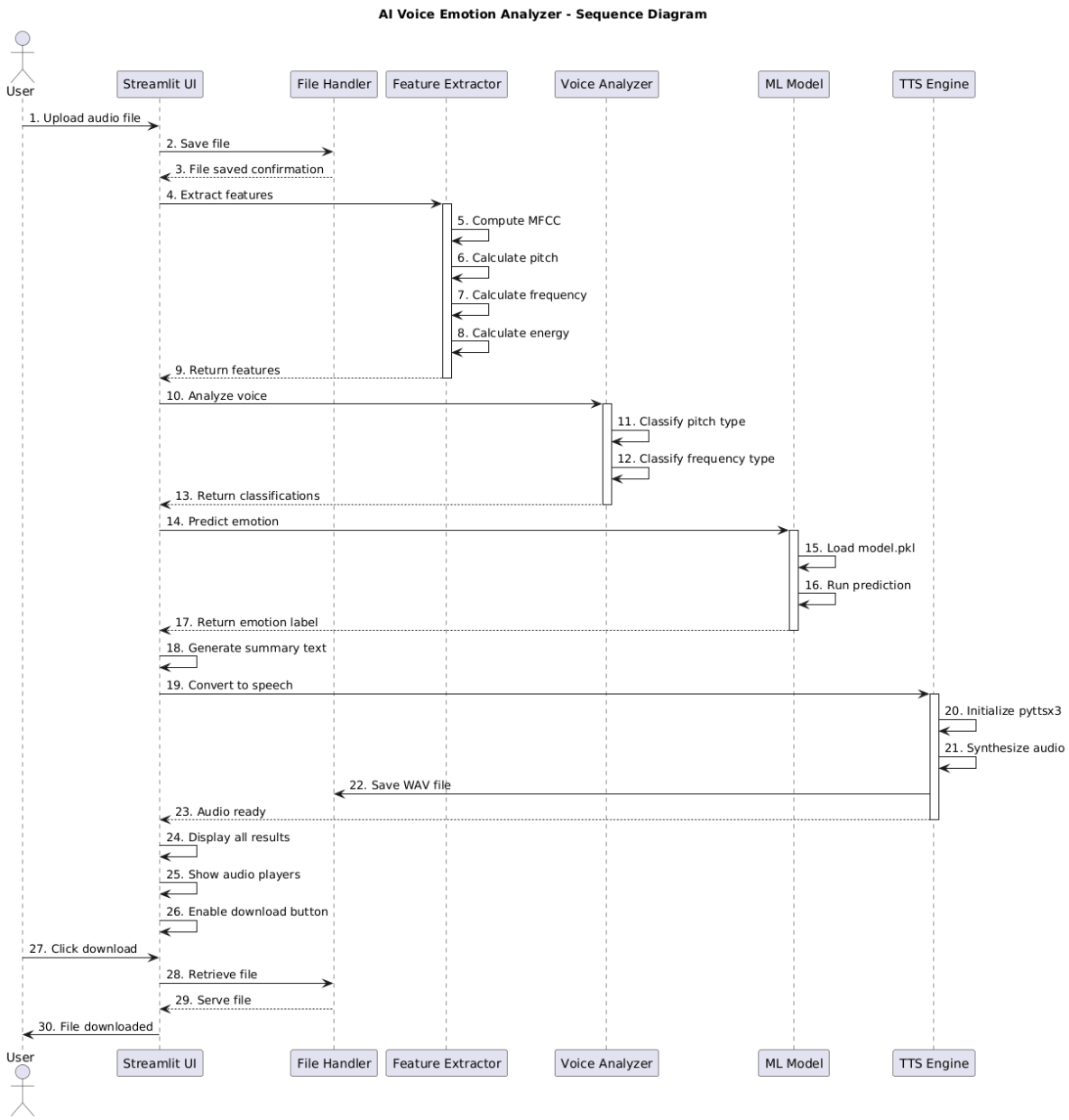


Project Structure

```
AI-Voice-Analyzer/
│
├── data/
│   └── ravdess/            # Dataset
│
├── audio/                  # Uploaded files & Generated output
│
├── src/
│   ├── feature_extractor.py  # Feature extraction logic
│   ├── dataset.py            # Dataset loader
│   ├── train.py              # Model training
│   ├── predict.py            # Prediction logic
│   ├── analyzer.py           # Pitch & emotion logic
│   ├── tts.py                # Text-to-Speech generation (NEW)
│   └── recorder.py           # (optional mic input)
```

```
│
├── model.pkl              # Trained model
├── streamlit_app.py       # UI application
├── requirements.txt
└── README.md
```

Sequence Diagram



Dataset Used

The project uses the **RAVDESS** (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset.

Key Features:

- Contains audio recordings of different emotions
- Structured file naming format
- Multiple actors and voice variations
- 8 emotions captured (calm, happy, sad, angry, fearful, surprised, disgusted, neutral)

Feature Extraction

Feature extraction is done using the **Librosa** library.

Extracted Features:

1. **MFCC (Mel Frequency Cepstral Coefficients)**
    - Used for training the machine learning model
    - Captures audio patterns and speech characteristics
2. **Pitch**
    - Represents the fundamental frequency of the voice
    - Filtered to human range (50–400 Hz) for accuracy
    - Helps determine if voice is High or Low pitch
3. **Frequency (Spectral Centroid)**
    - Indicates brightness of sound
    - Used to classify High / Low frequency
4. **Energy**
    - Measures loudness of the signal
    - Helps understand intensity of speech

Machine Learning Model

The model used is:

Random Forest Classifier

- Works well with structured features (MFCC)
- Handles non-linear data
- Provides good accuracy (~90%)

Training Process:

1. Load dataset
2. Extract features from all audio files
3. Split into training and testing data
4. Train model
5. Save model as model.pkl

Text-to-Speech (TTS) Module

A new module has been added to generate spoken output of the analysis results.

Technology Used:

- **pyttsx3** - Offline TTS engine
- No internet required
- Supports multiple voices and speed control

Features:

- Converts analysis text to speech
- Saves output as .wav file
- Playable directly in the interface
- Downloadable for users

Generated Speech Format:

"The emotion is calm. The pitch is high pitch and the frequency is high."

Voice Analysis Logic

After feature extraction:

- **Pitch** is categorized as:
    - High Pitch (> 200 Hz)
    - Low Pitch (≤ 200 Hz)
- **Frequency** is categorized as:
    - High Frequency (> 2000 Hz)
    - Low Frequency (≤ 2000 Hz)
- **Emotion** is predicted using the trained model

User Interface

The UI is built using:

Streamlit

Features:

- Upload .wav / .mp3 files
- Audio playback of original file
- Real-time analysis
- **Playback of generated TTS audio**
- **Download button for synthesized result**
- Clean and simple layout

Output Example

Emotion: Calm

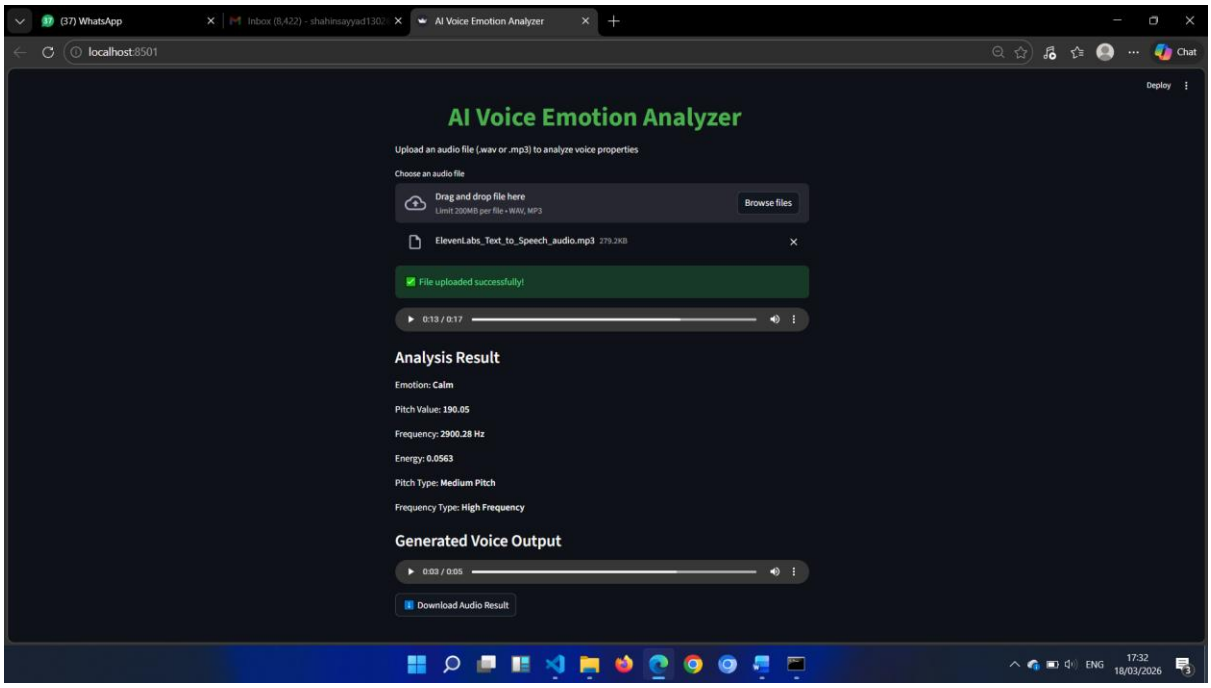
Pitch Value: 180.45

Frequency: 2100.32 Hz

Energy: 0.06

Pitch Type: High Pitch

Frequency Type: High Frequency

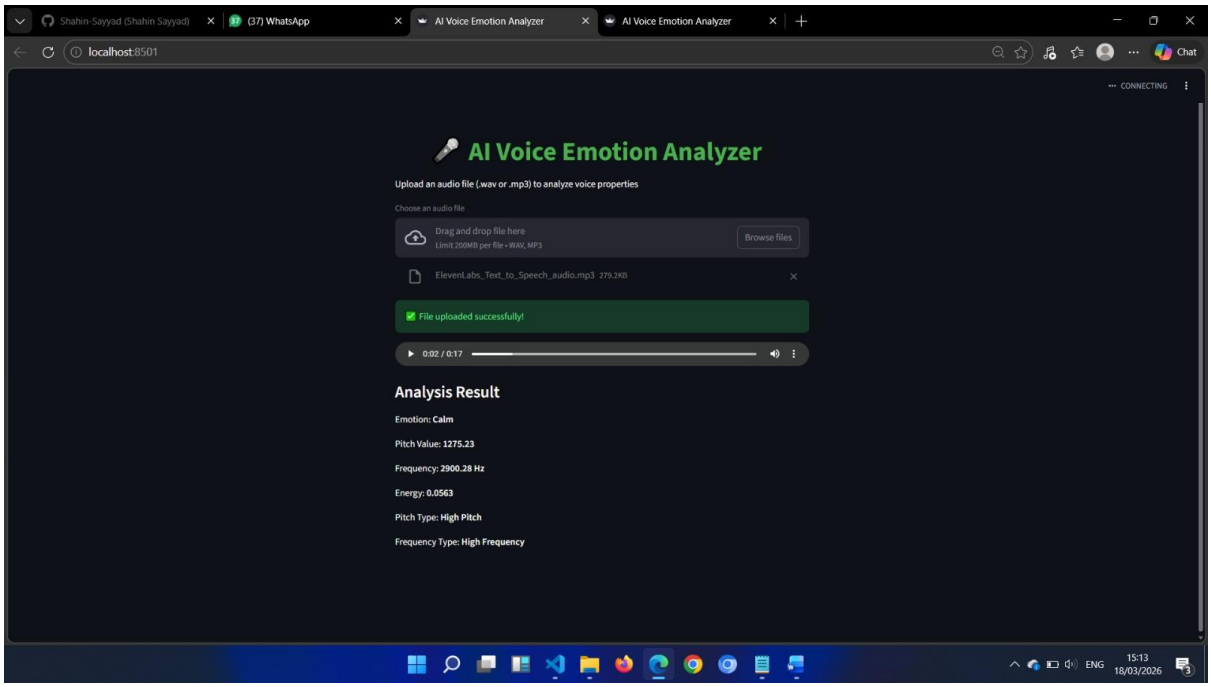


Generated Voice Output

[Audio Player] [Download Button]

Challenges Faced

- Incorrect pitch detection due to noise
- Dataset structure handling
- Import errors in Python modules
- Converting real-time input to file-based input
- **TTS engine initialization on different operating systems**
- **Cross-platform compatibility for audio generation**



Improvements Made

- Filtered pitch range for realistic values
- Improved feature extraction stability
- Added file upload support
- Integrated ML model for emotion prediction
- **Added TTS module for voice output generation**
- **Implemented download functionality**
- Enhanced UI for better presentation

Future Enhancements

- Real-time microphone input with instant TTS feedback
- Deep learning models (CNN / LSTM) for better accuracy
- **Emotional TTS (voice modulation based on detected emotion)**
- Multilingual voice support
- Graph visualization (waveform, pitch curve)
- Integration with APIs
- **Voice cloning for personalized output**

Conclusion

The AI Voice Emotion Analyzer successfully demonstrates how audio signals can be processed and analyzed using machine learning techniques. **The addition of Text-to-Speech output makes the system more interactive and accessible**, allowing users to hear the analysis results in natural language.

It provides both:

- **Technical output** (pitch, frequency, energy)
- **Human-understandable insights** (emotion, voice type)
- **Audible feedback** (synthesized speech results)

The system is scalable and can be enhanced further with advanced models, emotional TTS, and real-time processing capabilities.